MOHAMMED VI
FACULTY OF MEDICINE
UM6SS - CASABLANCA

Parathyroid SEMIOLOGY

Pr. IOANA HRISTOV

Semiology Endocrinology Module

1st Semester
Academic year: 2026-2027

MOHAMMED VI UNIVERSITY
OF SCIENCES AND HEALTH
UM6SS

www.um6ss.ma

Headlines

I. Physiology reminder
II. Hyperparathyroidism
    1. Clinical Presentation
    2. Investigations
    3. Etiologies
III. Hypoparathyroidism
    1. Clinical Presentation
    2. Investigations
    3. Etiologies

1. Learning Objectives

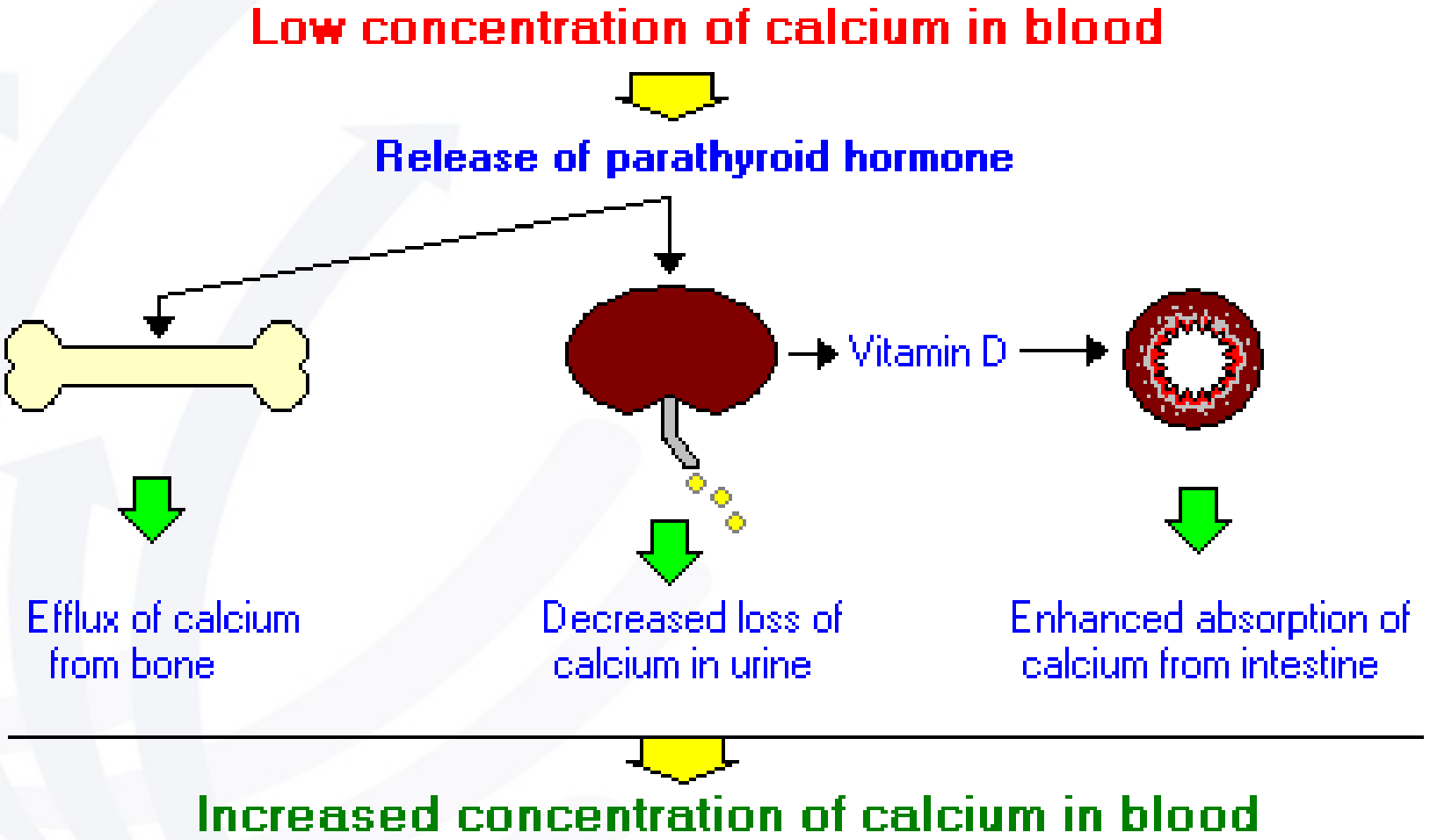
1. PRIMARY HYPERPARATHYROIDISM
2. SECONDARY HYPERPARATHYROIDISM
3. HYPOPARATHYROIDISM

Objectives

- Explain the clinical and biological manifestations of the main parathyroid disorders

- List the main etiologies of these different disorders

PARATHYROID HORMONE- PTH PHYSIOLOGY



Hyperparathyroidism

Usual onset is age 50-60

X3 more common in females

Lifetime incidence of 1 in 80 (>1%)

Classification

1) Primary

-Autonomous abnormal secretion (Adenoma)

-Hyperplasia (all 4 glands enlarged) (MEN syndrome?)

-Parathyroid Cancer is very rare

2) Secondary (Physiological)

-Renal failure

-Vitamin D deficiency

3) Tertiary

-Hyperplasia from long-term secondary hyperparathyroidism



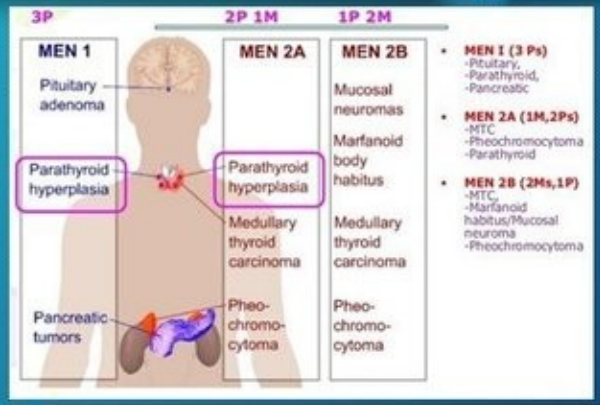


TABLE 3

Lab Comparison

Hyperparathyroidism	Calcium	PTH	Vitamin D	Phosphate
Primary	↑	↑→	↑	↓
Secondary	↓→	↑	↓	↑ or ↓
Tertiary	↑	↑↑	↓	↑

Key: ↑Elevated, ↓decreased, →normal.

Source: Brashers. *Pathophysiology.* 2015.[6]

INTRODUCTION

- Inappropriate and excessive secretion of PTH by one or more parathyroid glands.

•**Primary hyperparathyroidism:** results from a primary pathology of one or more parathyroids:

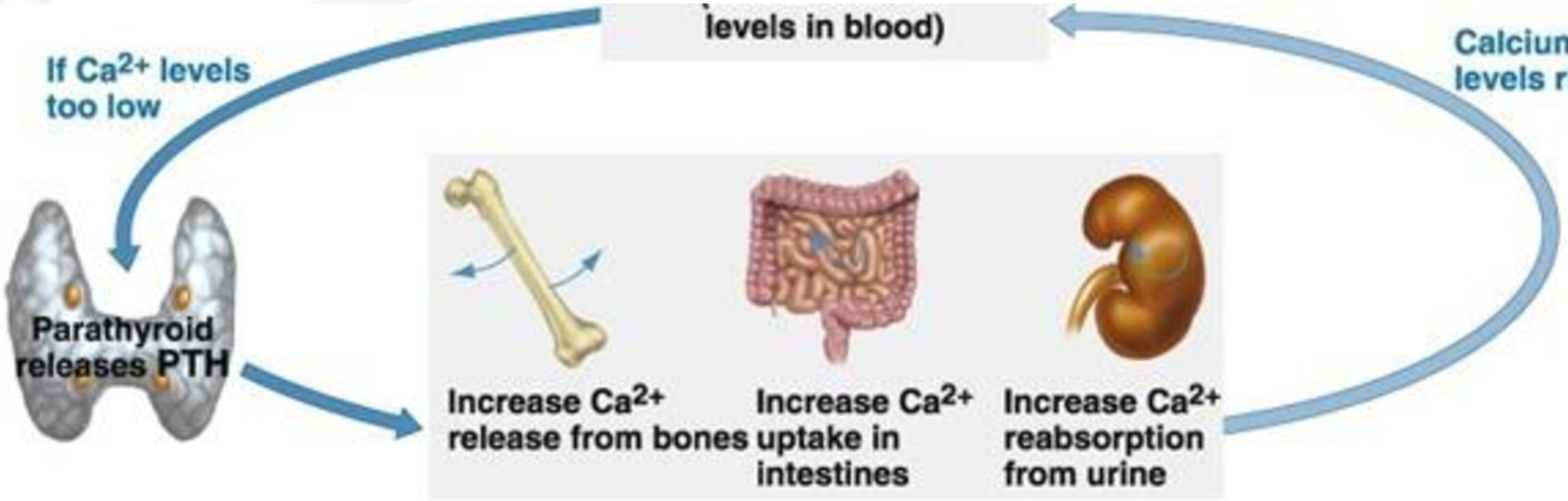
- **Adenoma (90%)**
- **Hyperplasia of one or more glands (10%)**
- **Carcinoma (exceptionally)**

• Asymptomatic+++.

•Diagnosis: Laboratoy tests

•Localization Studies++.

Primary hyperparathyroidism



PRIMARY HYPERPARATHYROIDISM

Bone resorption

Tubular reabsorption

Activation of vitamin D

Intestine absorption

→ **HYPERCALCEMIA**

CLINICAL FEATURES

1-Clinical presentation :

In descending order:

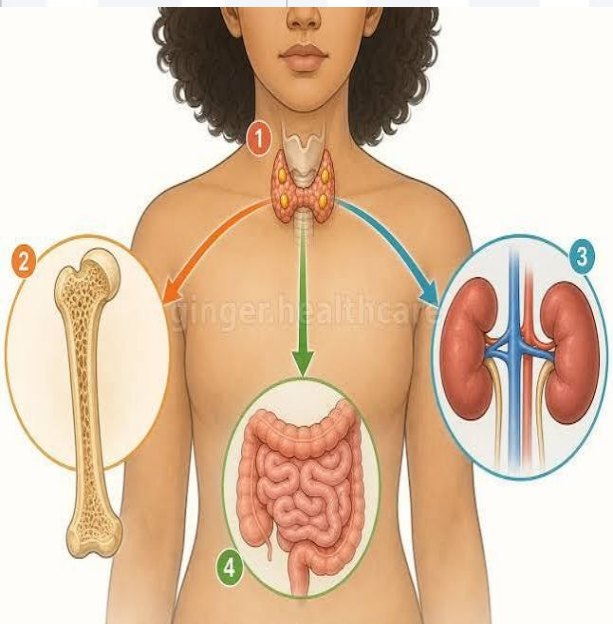
- Incidental finding in the presence of biological hypercalcemia
- Renal manifestations: kidney stones, renal impairment
- Bone manifestations (bone pain, pathologic fractures)
- Symptoms related to hypercalcemia (digestive, CNS, cardiovascular,…).

CLINICAL FEATURES

2-Clinical manifestations

2.1. Hyperparathyroid Kidney Disease

- Polydipsia and polyuria
- Renal colic.
- Kidney stone:
  - ➢ Multiple
  - ➢ Bilateral and recurrent
  - ➢ Nephrocalcinosis
  - ➢ Renal impairment (GFR<60ml/min)



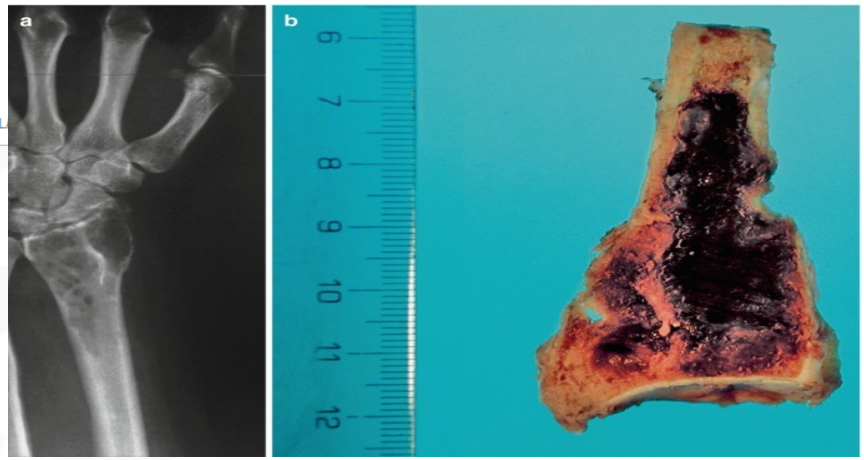
FACULTE DE MÉDECINE - UM6SS- CASABLANCA

CLINICAL FEATURES

2-Clinical manifestations

2.2 Bone Disease

- Bone pain: spine, pelvis, ribs
- Osteoporosis Fracture pathology
- Bone swellings in the jaw and long bones of the limbs are referred to as brown tumors, which are actually a proliferation of osteoclasts
- Late in life, there were settlements of the waist, skeletal deformations.



SUMMARY CLINICAL CONSEQUENCES OF HTPTH

Consequences of Hyperparathyroidism

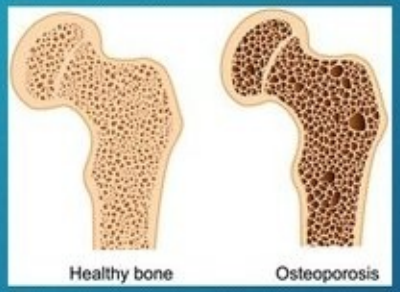
HYPERCALCAEMIA

- BONES (osteoporosis)
- MOANS (depression/confusion/reduced concentration)
- GROANS (abdominal pains, constipation, nausea, pancreatitis)
- STONES (renal stones)





- -Muscle weakness/Atrophy
- -Decreased neuroexcitation





- **Chronic Hypercalcaemia**

(Increased risk of stroke/heart failure/IHD)

(x2-3 Increased risk of several cancers (Breast/Colon/Kidney/Prostate)

(More deadly than thyroid cancer if left untreated)

Primary Hyperparathyroidism Workup

1. Laboratory tests

1.1- Diagnosis of PHP:

❑ Calcemia

- **Fasting sample of blood serum calcemia**
- **Verified hypercalcemia:**
  - ➢ controlled three times, three days in a row
  - ➢ Normal protein level
  - ➢ Adjusted to albumin ++++

Primary Hyperparathyroidism Workup

1. Laboratory tests

1.1- Diagnosis of PHP:

- **PTH i (1-84)**

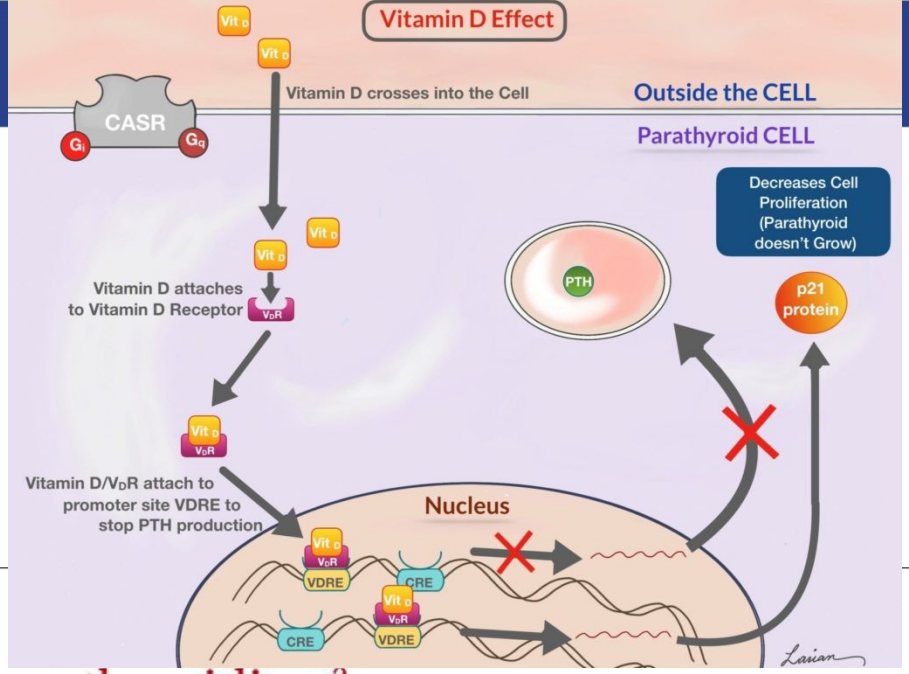
- High (80%) or normal inappropriate for hypercalcemia (20% cases).

Ca++ ionized↑↑+PTH ↑or Normal

=

hyperparathyroidism

DIFFERENTIAL DIAGNOSIS



TABLE

Differential diagnosis of primary hyperparathyroidism[2]

Diagnosis	Serum calcium	PTH	25(OH) vitamin D	Serum phosphorous
PHPT	High/Normal	High/Normal	Normal	Low/Normal
FHH	High/Normal	Normal/Mildly high	Normal	Normal/Low
Vitamin D deficiency	Low/Normal	High/Normal	Low	Normal/Low
Humoral hypercalcemia of malignancy	High	Low	Normal	Normal/Low
Milk alkali syndrome	High	Low	Normal	Normal/High

FHH, familial hypocalciuric hypercalcemia; PHPT, primary hyperparathyroidism; PTH, parathyroid hormone.

Primary Hyperparathyroidism Workup

Primary hyperparathyroidism:

- HYPERPARATHYROIDISM
- HYPERCALCEMIA
- HYPOPHOSPHATEMIA
- HYPERCALCIURIA

PTH ↑

Calcium ↑

Phosphorus ↓

Calciuria ↑

FACULTE DE MÉDECINE - UM6SS- CASABLANCA

Primary Hyperparathyroidism Workup

2.1- Cervical ultrasound:

- oval, anechoic or hypoechoic masses located posterior to the thyroid gland.

2.2 - Cervical T.D.M:

- Ectopic location: mediastinal

2.3- MRI:

- Difficult cases

2.4- 99mTc-sestamibi scintigraphy scanning : methoxy-isobutyl-isonitrile

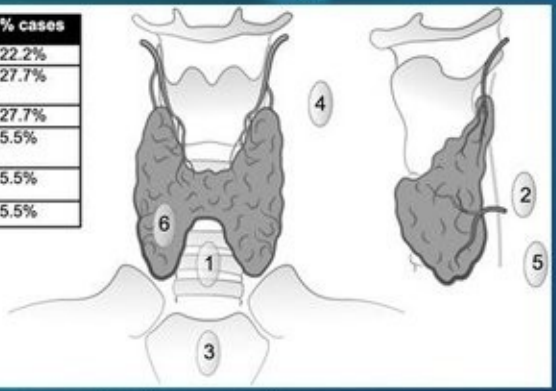
- Visualization of ectopic adenoma

Parathyroid Adenomas

- Benign tumour (over producing PTH)
- Can be the size of a grape
- 70% - one adenoma, 30% - >1 (5% have enlargement of all 4 glands)
- 10% are ectopic



	Anatomical location	% cases
1	Retrotracheal	22.2%
2	Deep tracheo-oesophageal groove	27.7%
3	Retrosternal	27.7%
4	Posterior to common carotid artery	5.5%
5	Retro-oesophageal	5.5%
6	Obscured by MNG	5.5%



Diagnosis:

-Often incidental hypercalcaemia

-Raised PTH on bloods (often 50-150)

-Renal stones/Bone #'s

Imaging?

-Sestamibi parathyroid scintigraphy

-USS (user dependant)

(Reveals presence/location of parathyroid tissue)



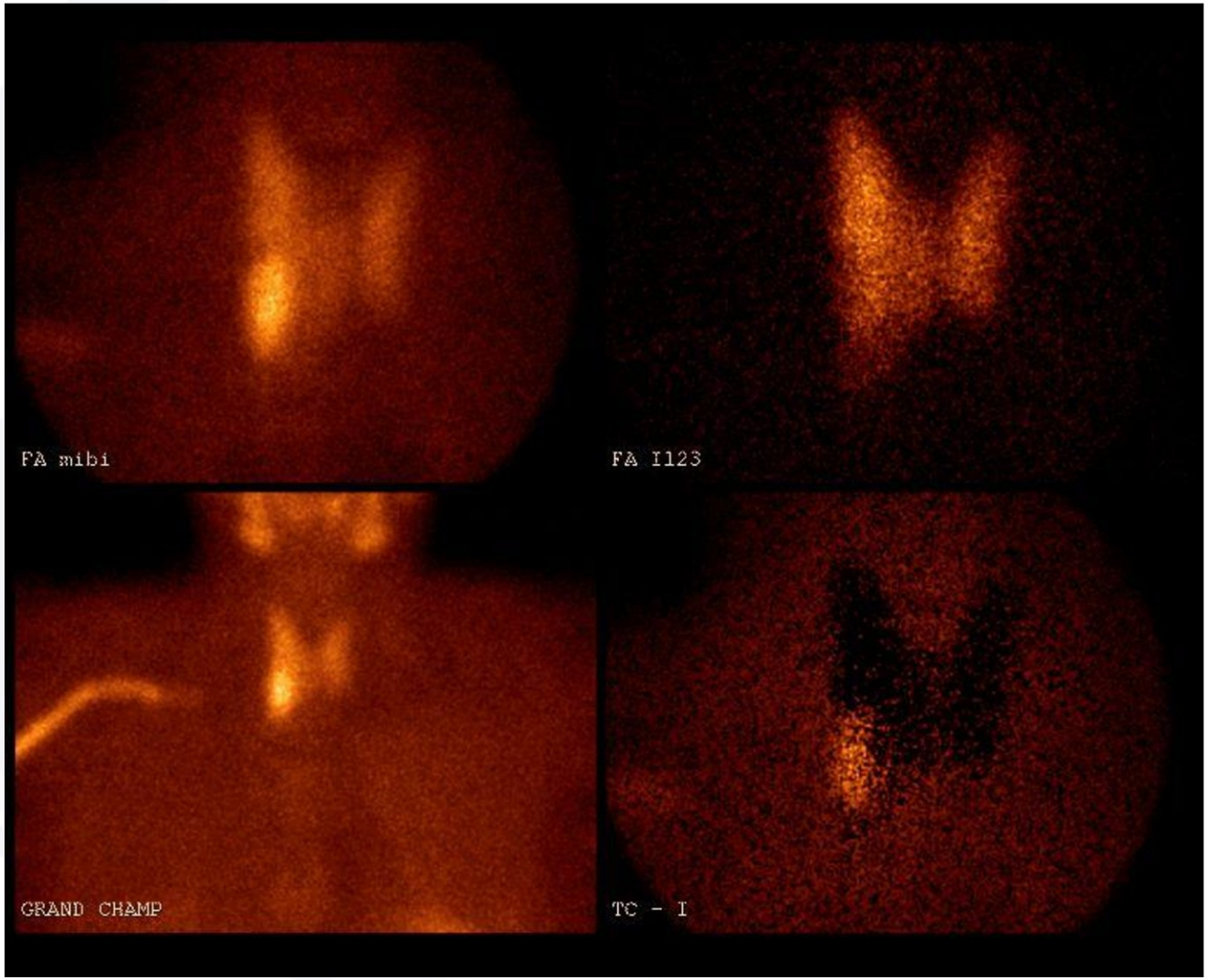


Treatment

-Parathyroidectomy (some normal thyroid tissue must be left in place)

Primary Hyperparathyroidism Workup

2. Localization Studies :



Clinical case

24 y.o woman

Superior right maxillary tumor
History of renal lithiasis
Frequent abdominal pain



Blood tests:

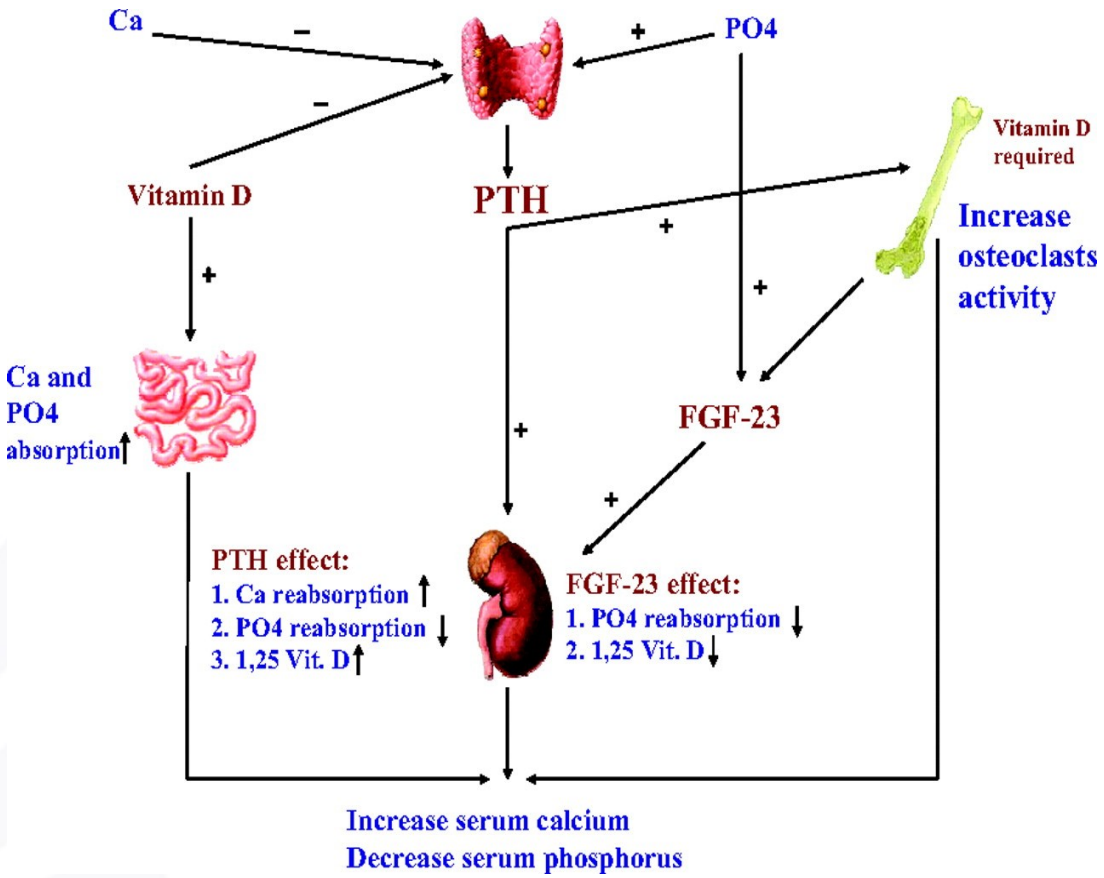
Calcemia=12.0 mg/dl
PTH=2900 pg/ml
Phosphorus=1.8 mg/dl

1.Diagnosis suspicion?
2. Further diagnosis algorithm

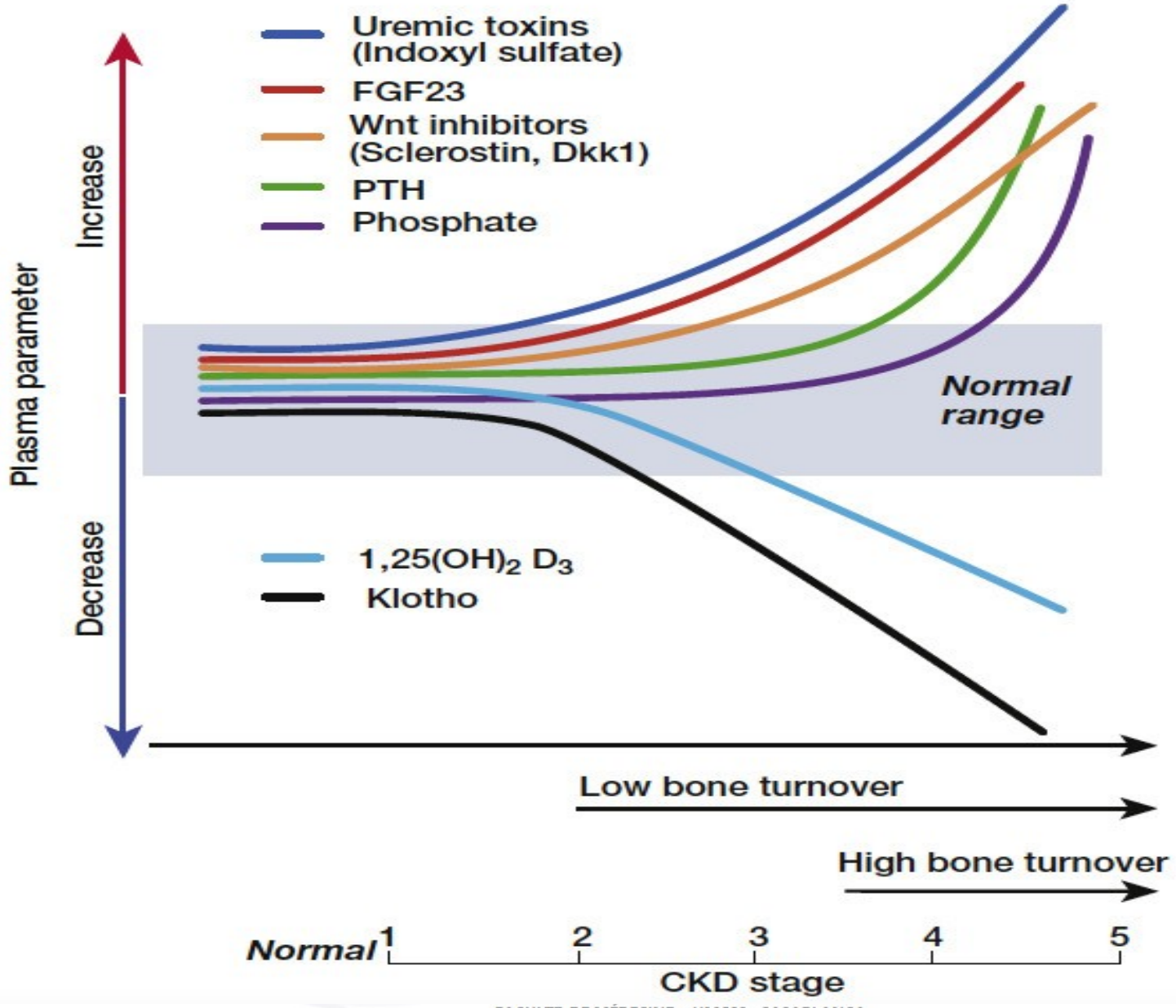
SECONDARY HYPERPARATHYROIDISM

- VITAMIN D DEFICIENCY

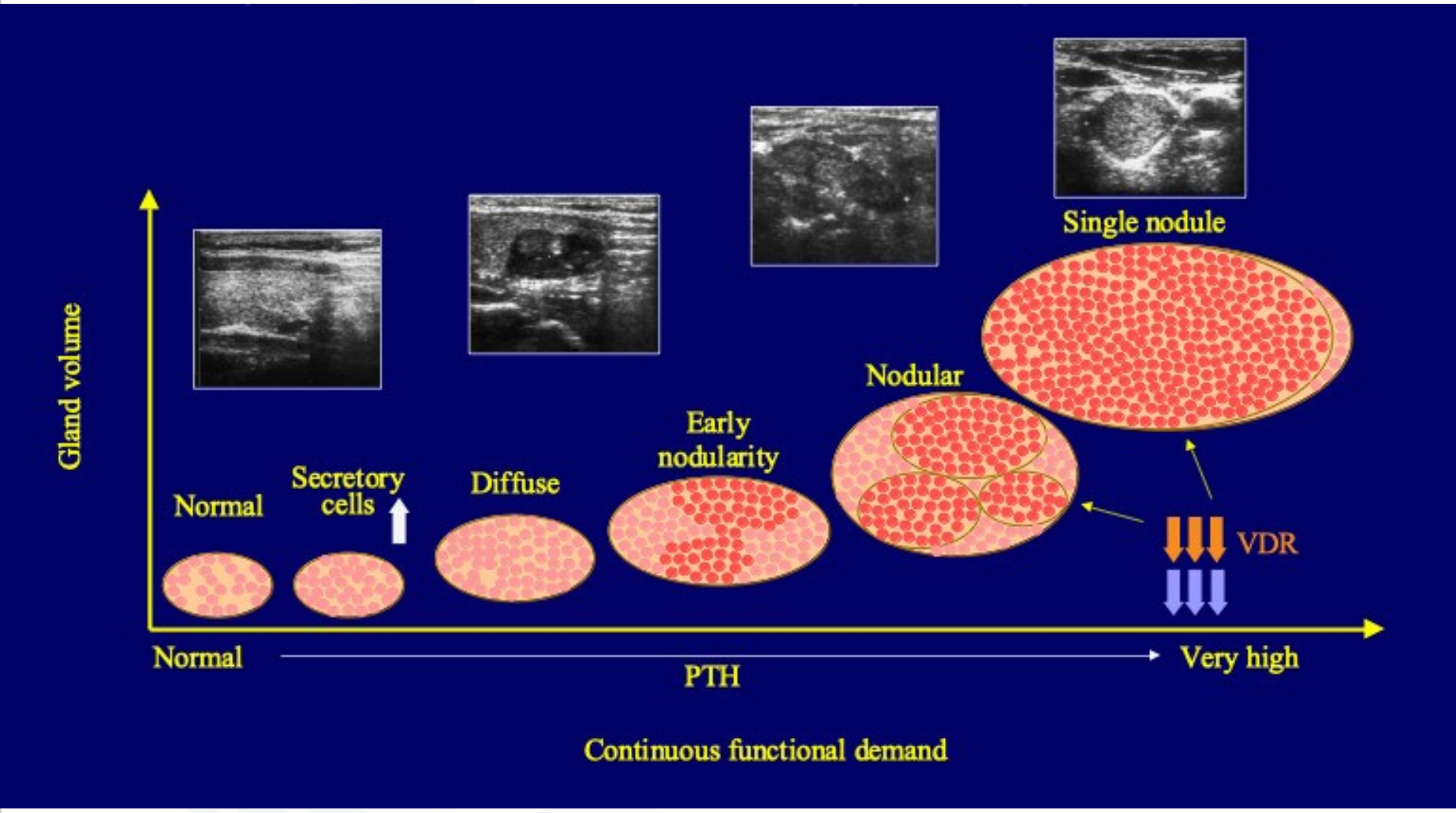
- END STAGE KIDNEY DISEASE



SECONDARY HPTH MECHANISM



Parathyroid gland hyperplasia



CLINICAL MANIFESTATIONS: CALCIPHYLAXY LESSSIONS



Clinical case:

72 y.o male

History of CKD- hemodyalisis for 15 years

Phospho-calcic metabolism:

Calcemia= 10.2 mg/dl

PTH=1800 pg/ml

Phosphorus = 5.3 mg/dl



1. Diagnosis suspicion?
2. Further investigation for positive diagnosis?

HYPOPARATHYROIDISM

INTRODUCTION

- Results from the decrease or a lack of PTH secretion, which results in hypocalcemia and hyperphosphatemia
- Acquired, Familial.
- Life-threatening situation: tetanic attacks or seizures.

CLINICAL MANIFESTATIONS OF HYPOCALCEMIA

1. Neuromuscular +++

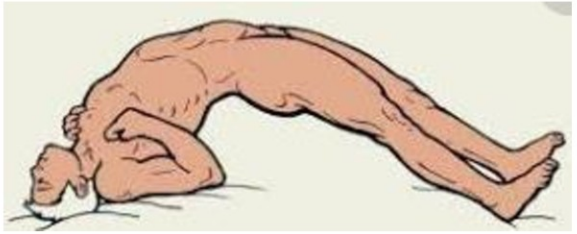
1.1 Tetany crisis



tonic and painful

contractions of the muscles

- **Onset:** Tingling and numbness of fingers, toes, or lips
- **- Muscle cramps:**
- **Carpopedal spasm:**
- Extension of the trunk in opisthotonos.



- **Painful but not painful crisis**
- **Resolves spontaneously in a few minutes**

CLINICAL MANIFESTATIONS OF HYPOCALCEMIA

1.neuromuscular +++

1.1 Tetany crisis

- **Association of smooth muscle damage:**

- Stridor due to laryngospasm
- Bronchial tubes -> bronchospasm
- Diaphragm breathing disorders
- Stomach, pylorus digestive disorders
- **Spontaneous or triggered by an effort**

CLINICAL MANIFESTATIONS OF HYPOCALCEMIA

2.2. Signs of latent tetany: Signs of neuromuscular excitability



ETIOLOGIES

1. Acquired :

➢ **Post-surgical +++**

- Thyroid

- Parathyroid

➢ **Other :**

- Granulomatous, neoplastic, inflammatory infiltration

2. Neonatal

3. Genetics

- Anomalies of the PTH gene

- Anomalies of parathyroid development

INVESTIGATIONS

1. Laboratory tests

- Hypocalcemia (↓ total calcium, ↓ ionized calcium)
- Hyperphosphatemia (because the phosphaturic effect of PTH is lost) and urinary phosphate ↓
- Low or undetectable PTH level
- Calciuria can be checked for low levels (< 100 mg/24 h) under normocalcium diet (600 to 1000 mg calcium)

Paraclinical

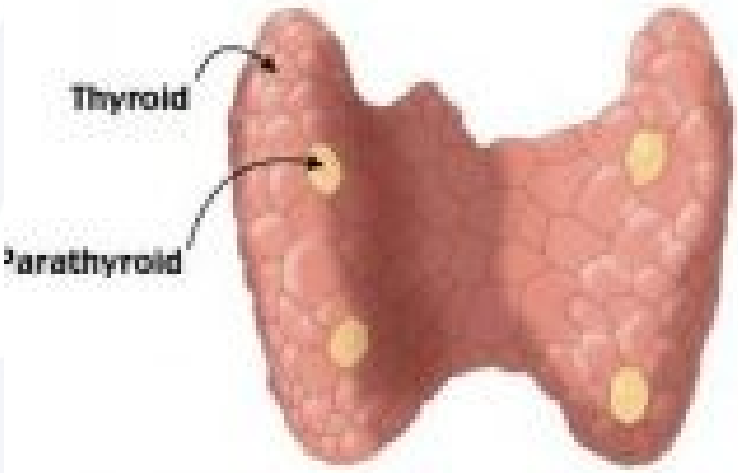
- Hypocalcemia
- Low or normal PTH: inappropriate
- Low Calciuria
- High phosphoremia levels

=

Hypoparathyroidism

Hypoparathyroidism

↓PTH = ↓Calcium



Causes

- Thyroid surgery
- Parathyroid surgery
- Autoimmune
- Infiltrative
- Familial
- Idiopathic

Hypocalcemia

- Tetany
- Chvostek sign (Contraction of facial muscles after tapping facial nerve)
- Trousseau sign (Induction of carpal pedal spasm)
- Paresthesias (Fingertips/perioral)
- Prolonged QT interval

1. References

1. Textbook Of Pathology By Harsh Mohan; 7th edition; Page no. 807 – 808.
2. Robbin's Basic Pathology; 5th edition; Page no. 1147 – 1148.

3. Fraser W. Bone and Mineral Metabolism. In: Rafai N, Horvath A, Wittwer C, editors. Tietz Textbook of Clinical Chemistry and Molecular Diagnostics. 6th Ed. St. Louis (MO): Elsevier; 2018. p.1422-1491.
4. Winter W, Harris N. Calcium Biology and Disorders. In: Clarke W, editor. Contemporary Practice in Clinical Chemistry. 2nd Ed. Washington DC: AACC Press; 2011. p. 505-526.